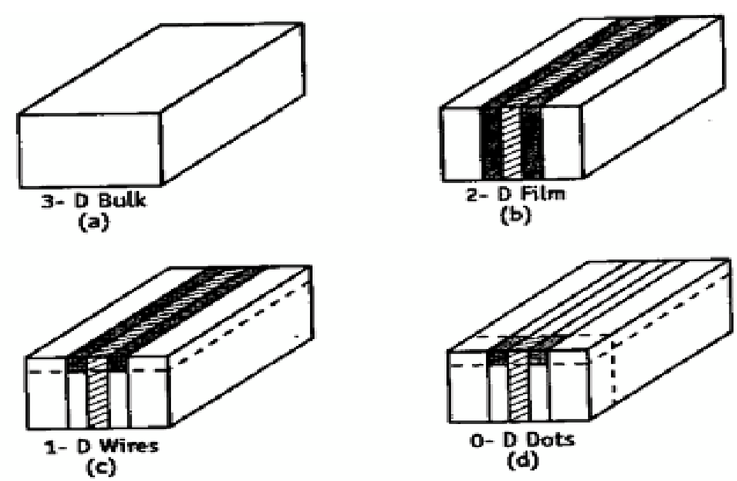
UNIT-IV
NANOTECHNOLOGY

Introduction

In 1959, Richard Feynman made a statement ‘there is plenty of room at the bottom’. Based on his study he manipulated smaller units of matter. He prophesied that “we can arrange the atoms the way we want, the very atoms, all the way down”. The term ‘nanotechnology’ was coined by Norio Taniguchi at the University of Tokyo. Nano means $10^{-9.}$ A nano metre is one thousand millionth of a metre (i.e. 10^{-9} m).
Nanomaterials could be defined as those materials which have structured components with size less than 100nm at least in one dimension. Any bulk material we take, its size can express in 3-dimensions. Any planer material, its area can be expressed in 2-dimension. Any linear material, its length can be expressed in 1-dimension.



SCHEMATIC DIAGRAM OF QUANTUM STRUCTURES

Materials that are nano scale in 1-dimension or layers, such as thin films or surface coatings.
Materials that are nano scale in 2-dimensions include nanowires and nanotubes.
Materials that are nano scale in 3- dimensions are particles like precipitates, colloids and quantum dots.

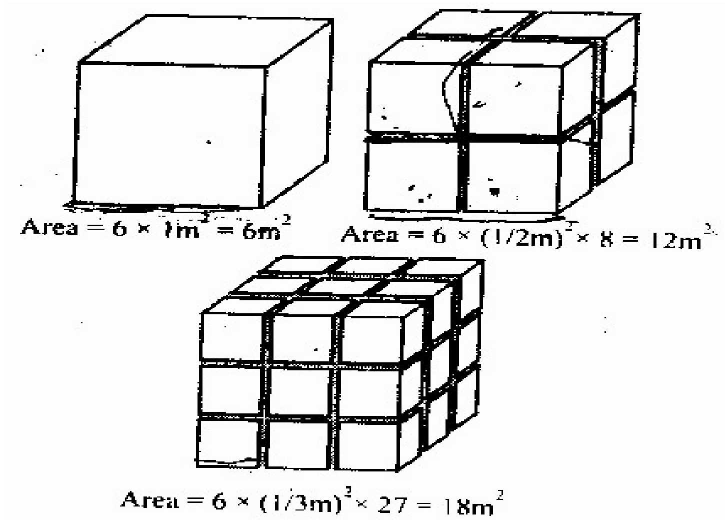
Nanoscience: it can be defined as the study of phenomena and manipulation of materials at atomic, molecular and macromolecular scales, where properties differ significantly from those at a larger scale.

Nanotechnology: It can be defined as the design, characterization, production and application of structures, devices and systems by controlling shape and size at the nano metre scale. It is also defined as “A branch of engineering that deals with the design and manufacture of extremely small electronic circuits and mechanical devices built at molecular level of matter. Now nanotechnology crosses and unites academic fields such as Physics, Chemistry and

Computer science.

4.1 Increase in surface area to volume ratio

Nano materials have relatively larger surface area when compared to the volume of the bulk material.



Consider a sphere of radius r
Its surface area $=4\Pi r^2$
Its volume $= 4\Pi r^3/3$

$$\frac{\text{Surface area}}{\text{Volume}} = \frac{4\Pi r^2}{4\Pi r^3/3} = \frac{3}{r}$$

Thus when the radius of sphere decreases, its surface area to volume ratio increases.

EX: For a cubic volume,

Surface area $= 6x1m^2 = 6m^2$

When it is divided it 8 pieces
It surface area $= 6x\ (1/2m)^{2x8} = 12m^2$
When the sane volume id divided into 27 pieces,

It surface area $= 6x\ (1/3m)^{2x27} = 18m^2$
Therefore, when the given volume is divided into smaller pieces, the surface area increases. Hence as particle size decreases, greater proportions of atoms are found at the surface compared to those inside. Thus nano particles have much greater surface to volume ratio. It makes material more chemically reactive.

As growth and catalytic chemical reaction occur at surfaces, then given mass of material in nano particulate form will be much more reactive than the same mass of bulk material. This affects there strength or electrical properties.

4.2 Quantum confinement effects

When the size or dimension of a material is continuously reduced from a large or macroscope size (such as centimeter) to a very small size, then it is observed that the properties of the matter remains same in the beginning but small changes occur.

afterwards. If the size drops below 100 nm (in nano range) then drastic changes occur in the properties. A progressive generation of diminishing size of rectangular nanostructures takes place in the following way.

If one dimension is reduced to the nano range while the third remains the same, then the structure so formed is known as quantum well as in fig (b).

If the two dimensions are reduced to nan orange while the third remains the same, then the structure so formed is known as quantum wire as shown in fig (1).

When all the 3-dimensions of the material are reduced to nano-range, then it is called as quantum dot as shown in fig (d).

It is important to mention here that the word quantum is associated with all the three structures. This is because the changes in properties arise from quantum mechanical nature.

When the dimensions of potential box is order of de-Broglie wave length of electron, the energy levels of electrons change. This effect is called Quantum configuration.

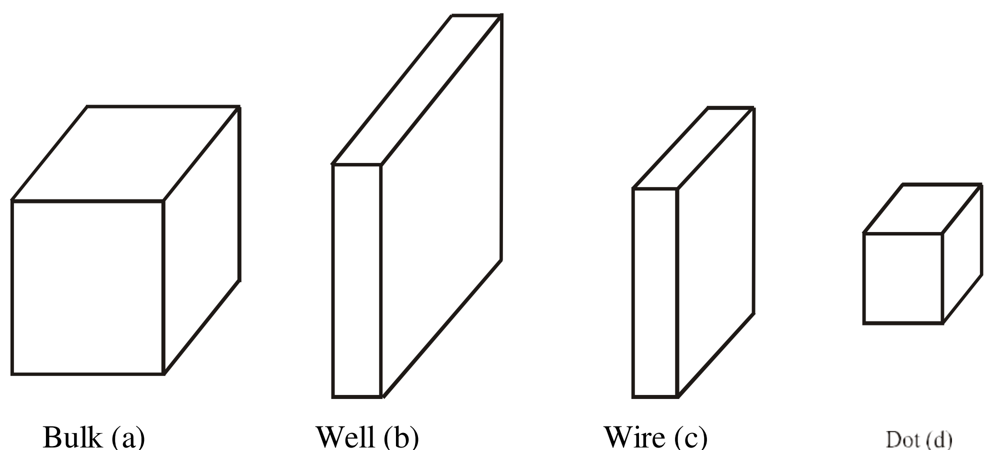


Fig: Progressive generation of rectangular nano structures

4.3 Synthesis of nanoparticles

The following two main techniques are used for the preparation of quantum dots.

1. Bottom-up Technique

2. Top-down Technique

Bottom-up Technique: This is a technique in which materials and devices are built up atom by atom i.e, technique to collect, consolidate and fashion of individual atoms and molecules into the structure. There is carried out by a sequence of chemical reactions controlled by series of catalysts. This process is used widely in biology. For examples, catalysts called enzymes assemble amino acids to construct living tissues that forms and supports the organs of the body.

Top-Down Technique: This is a technique in which materials are synthesized or constructed by removing existing materials from larger entities. Therefore in this technique, a large scale object or pattern is gradually reduced in dimensions or dimensions to mono scale pattern. This can be accomplished by a technique called lithography. Lithography is an image that is produced by making a pattern on the stone, inking the stone and then pushing the inked stone onto the paper. The lithography used may be a nano scale lithography of dip-pen lithography or an E-bean lithography. The lithography shines radiation through a tip to the surface coated with radiation - sensitive resist. The resist is then removed and the surface is chemically treated to produce the nanostructure.

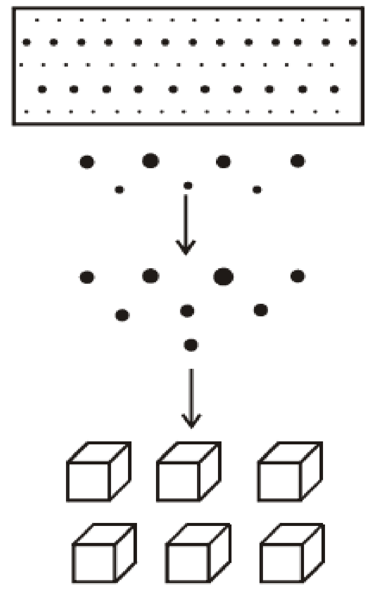


Fig: Nano particles

Nano materials can be synthesized by two techniques.

Bottom-up technique: In this technique, the nano structured materials are synthesized by assembling the atoms or molecules together to form the nano materials.

- Sol-Gel method
- Precipitation method
- Combustion method

Top-Down technique: In this technique, the bulk solids are dis-assembled (broken) into finer pieces until the particles are in the order of nanometer.

- Ball milling (Mechanical Griding) method
- Chemical vapour deposition (CVD) method
- Physical vapour deposition (PVD) method

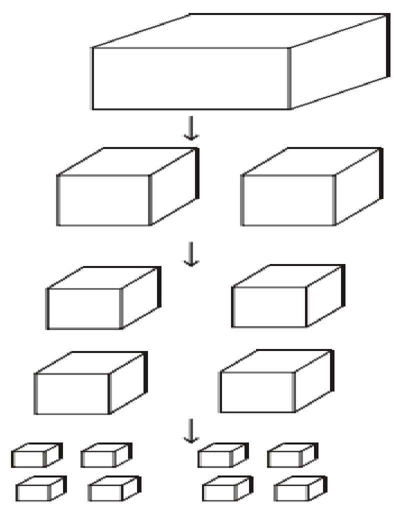


Fig.: Nano Structure

4.3.1 Sol-Gel Method

Sol-Gel is a chemical process used to make ceramic and glass materials in the form of thin films, fibers or powders.

A sol is a colloidal or molecular suspension of solid particles of ions in a solvent and a Gel is a semi-rigid mass that forms when the solvent form the sol begins to evaporate.

The precursors for synthesizing the colloids are metal alkoxides and metal chlorides [Ex: Tetramethoxy (TMOs) and Tetra ethoxy, silane (TEOs)]. The starting material is processed with water or dilute acid in an alkaline solvent. The material undergoes hydrolysis and poly condensation reaction which leads to the form of colloids. The colloid system composed of solid particles dispersed in a solveril. The sol is then evolved to form an inorganic network containing liquid phase (gel). The schematic representation of the synthesis of nano particles using sol. Gel method is shown in below figure.

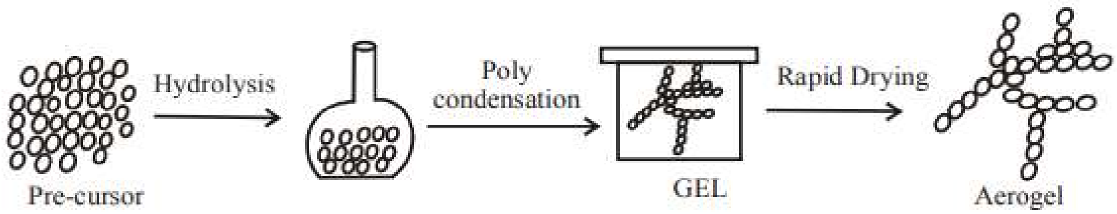


Fig.

Advantages:

- This method is low cost and low temperatures technique.
- One can get a highly porous material like glass and glass ceramics.
- One can get monetized nano particles.

Disadvantages:

- Controlling the growth of particle is different difficult.
- Slopping the newly formed particles from agglomeration is also difficult.

Note:

A precursor is a compound that participates in a chemical reaction and produces another compound.

Allotropy is the property of some chemical element to exist in two or more different forms in the same physical state.

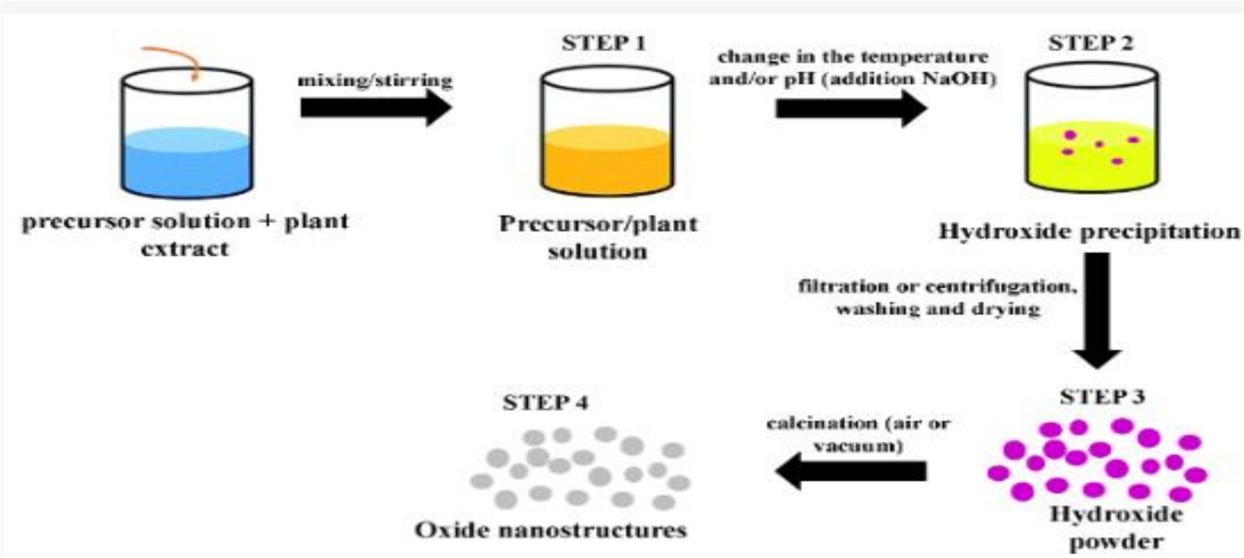
4.3.2 Chemical Precipitation and Co-precipitation

A chemical precipitation process consists of three main steps:

- chemical reaction
- nucleation
- crystal growth

Chemical precipitation is generally not a controlled process in terms of reaction kinetics and the solid phase nucleation and growth processes. Therefore, solids obtained by chemical precipitation have a wide particle size distribution plus uncontrolled particle morphology, along with agglomeration. To obtain nanoparticles with a narrow size distribution, the necessary requirements are (i) a high degree of supersaturation, (ii) a uniform spatial concentration distribution inside a reactor and (iii) a uniform growth time for all particles or crystals.

The other commonly used solution method for the synthesis of multi component oxide ceramics is co-precipitation method, which produces a "mixed" precipitate comprising two or more insoluble species that are simultaneously removed from solution. The precursors used in this method are mostly inorganic salts (nitrate, chloride, sulfate, etc.) that are dissolved in water or any other suitable medium to form a homogeneous solution with clusters of ions. The solution is then subjected to pH adjustment or evaporation to force those salts to precipitate as hydroxides, hydrous oxides, or oxalates. The crystal growth and their aggregation are influenced by the concentration of salt, temperature, the actual pH and the rate of pH change. After precipitation, the solid mass is collected, washed and gradually dried by heating to the boiling point of the medium.



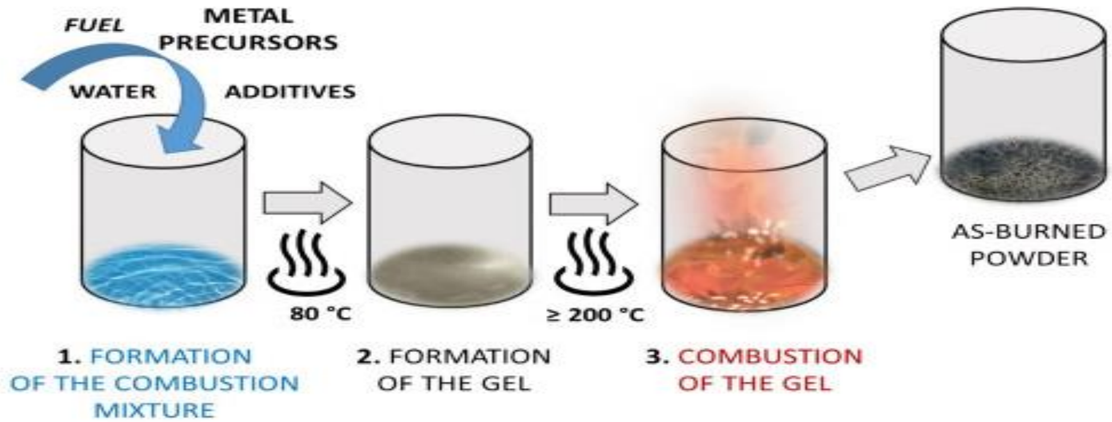
4.3.3 Combustion method

Combustion reaction: when a substance reacts with oxygen, forming light and heat in the form of fire.

Combustion is a high-temperature exothermic (heat releasing) redox (oxygen adding) chemical reaction between a fuel and an oxidant.

For example, when methane burns in oxygen, it releases carbon dioxide and water. The formation of carbon dioxide shows the carbon present in the methane has undergone oxidation.

For the combustion process, fuel and oxidizer are required; metal nitrate acts as an oxidizing reactant and urea acts as a reducing reactant. The synthesis process involves weighing solid precursors, then grinding and mixing generally in an aqueous solution of distilled water in which the ingredients are dissolved.



4.3.4 Chemical vapour deposition(CVD) method

In this method, nano particles are deposited from gas phase. Material is heated to form a gas and then allowed to deposit on an solid surface. Usually under high vacuum. In deposition by chemical reaction new products are formed.

In this method a metal organic precursor is introduced into the hot zone of the reactor using mass flow controller. The precursor is vaporized either by resistive or inductive healing. The carrier gas such as Ar (or) Ne carries the hot atoms to the reaction chamber. The hot atoms collide with cold atoms and undergo condensation through nucleation and form small clusters inside reaction chamber other reactants are added to control the reaction rate. Then these clusters are added to condense on a moving bell arrangement with scrapper to collect the nano particles. The particle size can be controlled by rate of evaporation, rate of cluster formation and rate of condensation (cluster removed from the reaction chamber).

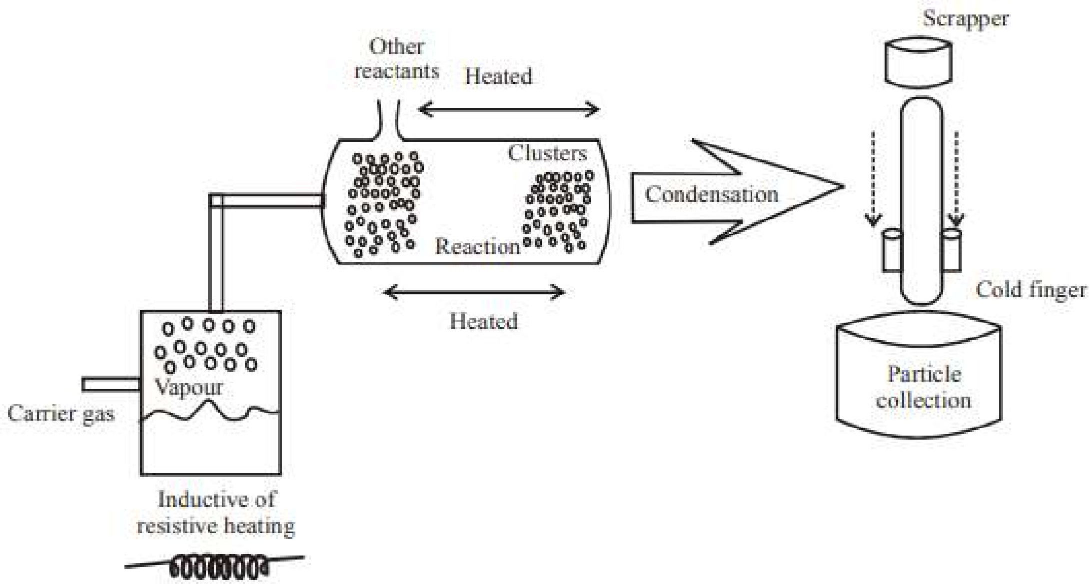


Fig.

Advantages:

- Wide range of ceramics including nitrides and carbides can be synthesized.
- More complex oxides such as BaTiO3 or composite structures can be formed.
- Controlling growth of particles is easy.

Disadvantages:

- High cost and high temperature technique.

4.3.5 Physical Vapour Deposition (PVD) method

In this method the experimental set up consists of a bell jar in which an inert gas or reactive gas is filled after vacuum. The materials to be evaporated are placed in the crucible and are heated either his resistance or an electron bombardment until sufficient vapour develops. The evaporated atoms (or) molecular are allowed condensing on a cold finger which is externally cooled by liquid N2. The nano particles on the cold finger is scraped by the scraper and then collected to the piston anvil through a funnel. The piston anvil is used to obtain the compacted nano powders. The desired purity of the nano powder is obtained since the evaporation is done at the vacuum chamber with the pressure of an inert or reactive gas. This method is more suitable for non- conductive or high melting materials.

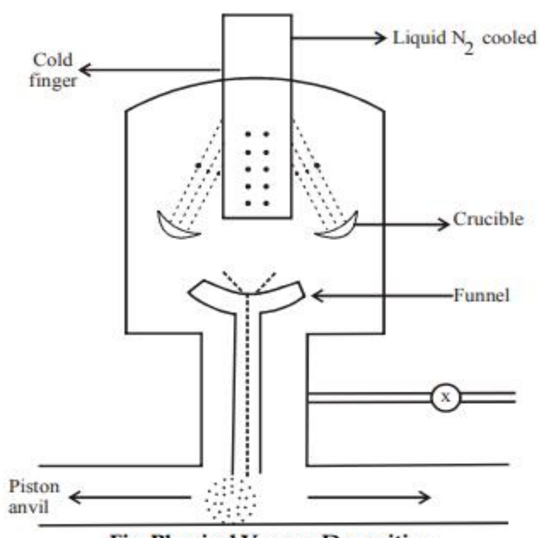


Fig. Physical Vapour Deposition

4.3.6 Ball milling method

In ball milling, also called mechanical crushing, small balls are allowed to rotate around the inside of a drum and then fall on a solid with gravity force and crush the solid into nano crystallites. Ball milling can be used to prepare a wide range of elemental and oxide powders. For example iron with grain sizes of 10-30 nm can be formed. Other crystallites such as iron nitriles can be made using ammonia gas. A variety of intermetallic compounds based on nickel and aluminum can be formed. Ball milling is the preferred method for preparing metal oxides.

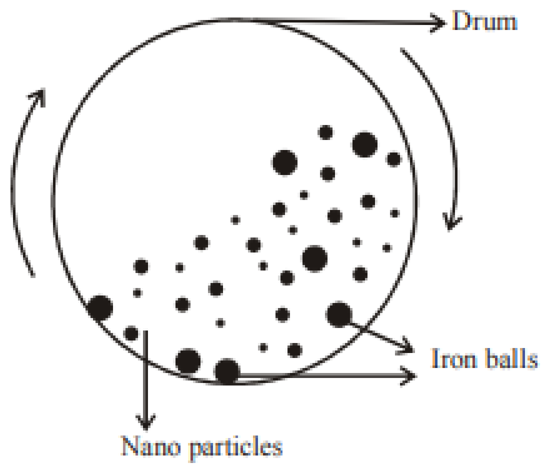


Fig. Ball Moulding

4.4. Characterization Techniques

4.4.1 Transmission Electron Microscope (TEM)Principle:

The transmission electron microscope works on the some basic principle as that of a light microscope except that an electron beam is used instead of light beam. Due to much lower wavelength associated with electrons. It is possible to get a resolution which is thousand times more in comparison as obtained with light microscope. Therefore, an enlarged version of the image of the specimen appears on the florescent screen (or) photographic plate. We can also consider the working principle of TEM likea slide projector.

Working: At the top of the microscope, there is electron gun which produces a stream of monochromatic electrons. The beam travels through vacuum int eh column of microscope.

The beam strikes the condenser lenses instead of glass lenses focusing, the microscope user electromagnetic lenses which focus the electrons into a very fine beam.

The beam strikes the specimen which we want to study. Depending on the density of the material present, some of the electrons are scattered and disappear from the beam. The other are transmitted through the specimen.

The transmitted electrons are focused by the objective lens into an image. The image is passed through the projector lenses.

The beam then hits a fluor's fluorescent screen. The screen shows a shadow like image of the specimen allowing the user to see the image.

It is observed that transmission of unscattered electrons is inversely proportional to the specimen thickness i.e., the thicker areas will have less transmitted light and thus will appear dark. On the other hand, the thinner areas will have more transmitted light and thus will appear brighter.

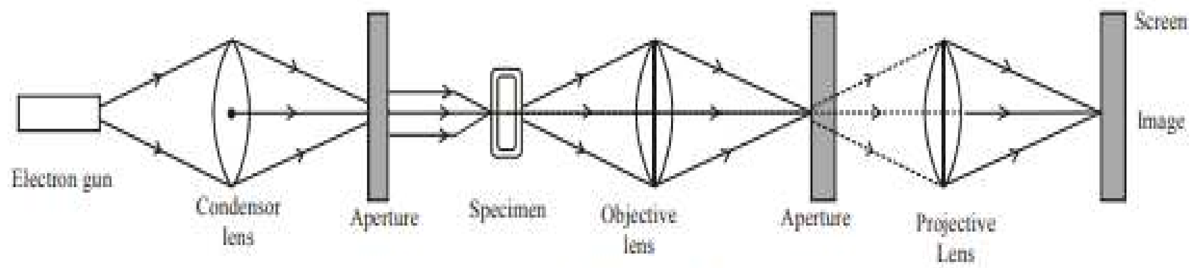


Fig. Transmission Electron Microscope

Uses:

The high magnification range and resolution has made the TEM as a valuable tool in medical, biological and material research.
The object of the order of 0.2 nm be viewed.
This allows the visualization of viruses and bacteria's.

4.4.2 Characterisation of Nano Particles by XRD

The analysis of crystalline materials semi crystalline materials and amorphous materials at the nano scale are possible with x-ray diffraction. XRD provides an adaptable method for measuring crystalline phases, degree of crystallinity and structures.
Measurements by XRD help to understand what really happens during processing. For characterization of nano particles X-ray differentiation (XRD) is the most widely used technique.
Typical XRD pattern of silver particles prepared by the chemical as shown in the below figure.
The X-RD study indicates the formation of silver (Ag) nano particles.
From this study considering the peak of diffraction angle at 450, average particle size has estimated by using Debye Scherrer formula.

$$D=(k\lambda/\beta \cos \theta)$$

'λ' is the wavelength of X-Ray (0.1541 nm)
' β ' is FWHM (full width at half maximum) (0.011 radian),
'θ' is the diffraction angle (450)
D is the particle diameter (size)
The average particle size is calculated to be around 14 nm.
Thus using XRD pattern we are able to calculate the particle size.
To 'see' the particle, we have to use TEM or SEM.

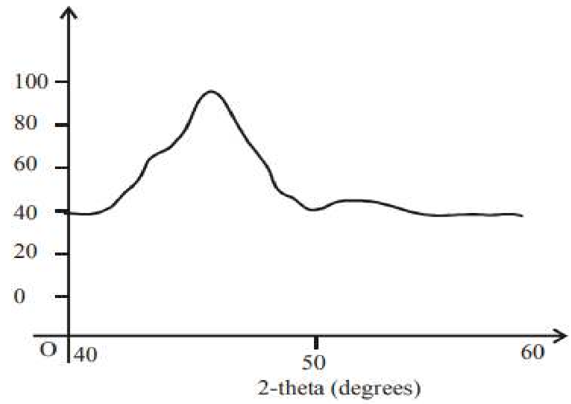


Fig.: X-ray diffraction of Ag nano particles

4.4.3 SCANNING ELECTRON MICROSCOPE (SEM)

The image is scanning electron microscopy (SEM) is produced by scanning the sample with a focused electron and detecting the secondary and / or back scattered electrons.
A schematic representation of a SEM is shown in the below figure.
The electron gun produces a stream of monochromatic electrons
The electron stream is condensed by the first condenser lens. It works in conjunction lens forms the electrons into a thin, light coherent beam.
The second condenser lens forms the electrons into a thin, light coherent beam.
Objective aperture further eliminates high angle electrons from the beam.
A set of coils acting as electrostatic lens scans and sweeps the beam in a grid fashion (as in television). The beam (dwells) on the points for a period of a time determined by the scan speed. Dwell time is usually in microsound range.
The objective lens focuses the scanning beam onto the part of the specimen.
When a beam strikes the sample interaction occurs. Before the beam moves to the next dwell point, the various instruments housed to measure various interactions count the number of interactions and display a pixel on a CRT.
The intensity of display is determined by the interaction number. More interactions give a brighter pixel.
This process is repeated, until the grid scan is finished and then repeated. The entire pattern can be scanned 30 times per second.

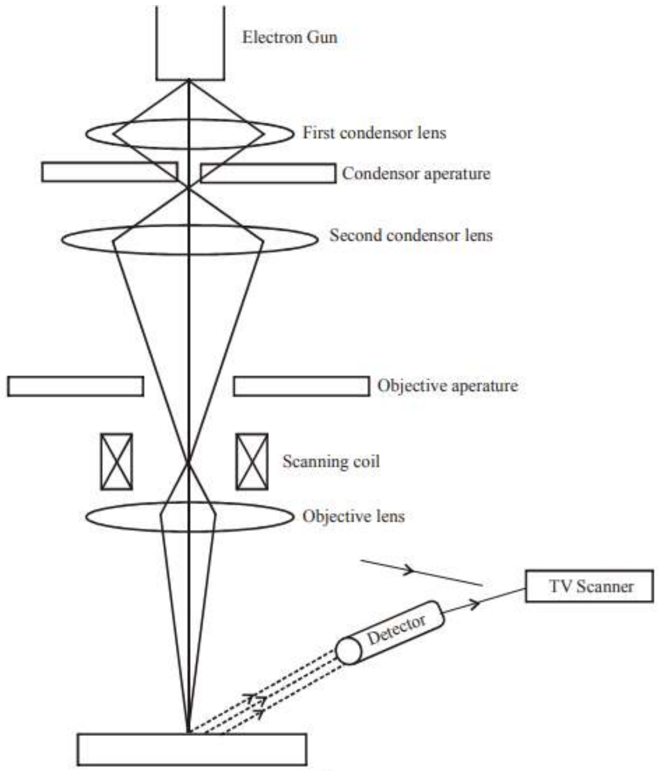


Fig.

Note: This method is only useful to single crystal particles of size more than 20 n.

4.5. DOMINANCE OF ELECTROMAGNETIC FORCES

Then particle size is downsized to nanoscale, gravitational forces become negligible and electromagnetic forces dominate.

Gravitational force between any two particles is a function of mass and distance, whereas electromagnetic force is a function of charge and distance and is independent of mass. Since the mass of a nanoscale object very small, the gravitational force is negligible. Since electromagnetic force is not influenced by the mass of the particles, there are strong electromagnetic forces between nanoscale particles. For example, the electromagnetic force between two protons is 1036 times stronger than the gravitational force.

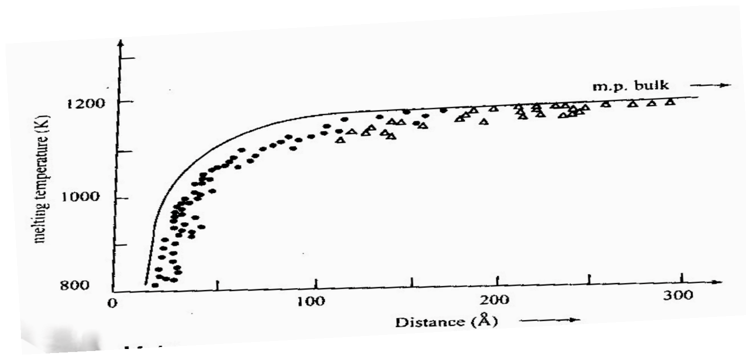
4.6. RANDOM MOLECULAR MOTION

Nano-sized materials, there exists random molecular motion which is also a size- dependent property. As long as the material is above absolute zero temperature, molecules in it move because of kinetic energy, this is called random molecular motion and it ceases at absolute zero. In case of bulk materials, this random motion of macroparticles is negligible as compared to the size of the particles. Hence, this random motion (which is also called the Brownian motion) is not affected by the motion of the particle. However, at nanolevel, these random motions are comparable with the size of the particles and influence the behaviour of the particles. The melting point of various nanomaterials with varying particle size will have different melting points. The melting point decreases with decrease in particle size.

4.7. Properties of nano materials

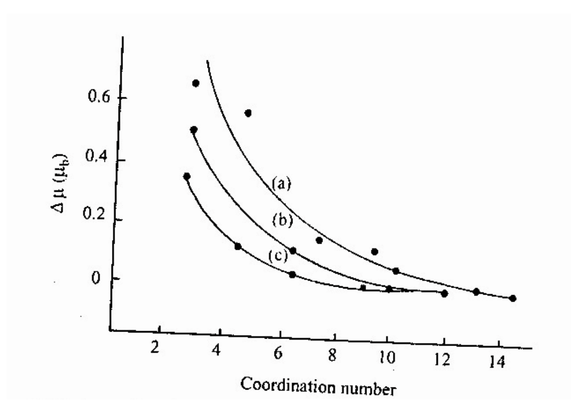
The physical, electronic, magnetic and chemical properties of materials depend on size. Small particles behave differently from those of individual atoms or bulk.

Physical properties: The effect of reducing the bulk into particle size is to create more surface sites i.e. to increase the surface to volume ratio. This changes the surface pressure and results in a change in the inter particle spacing. Thus the inter atomic spacing decreases with size.
The change in the inter particle spacing and the large surface to volume ratio in particle have a combined effect on material properties. Variation in the surface free energy changes the chemical potential. This affects the thermodynamic properties like melting point. The melting point decreases with size and at very small sizes the decrease is faster.



Chemical properties: the large surface to volume ratio, the variations in geometry and electronic structure has a strong effect on catalytic properties. The reactivity of small clusters increases rapidly even when the magnitude of the cluster size is changed only by a few atoms.

Another important application is hydrogen storage in metals. Most metals do not absorb, hydrogen is typically absorbed dissociatively on surfaces with hydrogen- to- metal atom ratio of one. This limit is significantly enhanced in small sizes. The small positively charged clusters of Ni, Pd and Pt and containing between 2 and 60 atoms decreases with increasing cluster size. This shows that small particles may be very useful in hydrogen storage devices in metals.



Electrical properties: The ionization potential at small sizes is higher than that for the bulk and show marked fluctuations as function of size. Due to quantum confinement the electronic bands in metals become narrower. The delocalized electronic states are transformed to more localized molecular bands and these bands can be altered by the passage of current through these materials or by the application of an electric field.

In nano ceramics and magnetic nano composites the electrical conductivity increases with reduction in particle size where as in metals, electrical conductivity decreases.

Optical properties: Depending on the particle size, different colours are same. Gold nano spheres of 100nm appear orange in colour while 50nm nano spheres appear green in colour. If semiconductor particles are made small enough, quantum effects come into play, which limits the energies at which electrons and holes can exist in the particles. As energy is related to wavelength or colour, the optical properties of the particles can be finely tuned depending on its size. Thus particles can be made to emit or absorb specific wavelength of light, merely by controlling their size.

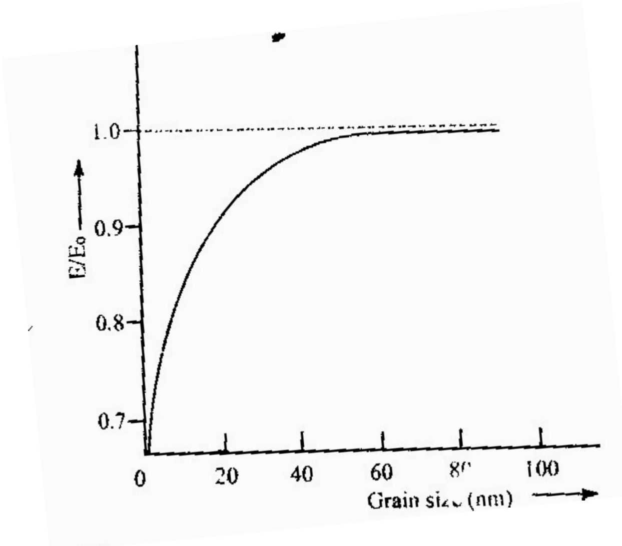
An electro chromic device consist of materials in which an optical absorption band can be introduced or existing band can be altered by the passage of current through the materials, or by the application of an electric field. They are similar to liquid crystal displays (LCD) commonly used in calculator and watches. The resolution, brightness and contrast of these devices depend on tungstic acid gel's grain size.

Magnetic properties: The strength of a magnet is measured in terms of coercivity and saturation magnetization values. These values increase with a decrease in the grain size and an increase in the specific surface area (surface area per unit volume) of the grains.

In small particle a large number or fraction of the atoms reside at the surface. These atoms have lower coordination number than the interior atoms. As the coordination number decreases, the moment increases towards the atomic value there is small particles are more magnetic than the bulk material.

Nano particle of even non magnetic solids are found to be magnetic. It has been found theoretically and experimentally that the magnetism special to small sizes and disappears in clusters. At small sizes, the clusters become spontaneously magnetic.

Mechanical properties: If the grains are nano scale in size, the interface area within the material greatly increases, which enhances its strength. Because of the nano size many mechanical properties like hardness, elastic modulus, fracture toughness, scratch resistance, fatigue strength are modified.



The presence of extrinsic defects such as pores and cracks may be responsible for low values of E (young's modulus) in nano crystalline materials. The intrinsic elastic modulli of nano structured materials are essentially the same as those for conventional grain size material until the grain size becomes very small. At lower grain size, the no. of atoms associated with the grain boundaries and triple junctions become very large. The hardness, strength and deformation behaviour of nano crystalline materials is unique and not yet well understood.

Super plasticity is the capability of some polycrystalline materials to exhibit very large texture deformations without fracture. Super plasticity has been observed occurs at somewhat low temperatures and at higher strain rates in nano crystalline material.

4.8 APPLICATION OF NANOMATERIALS:

1. **Engineering:** i).Wear protection for tools and machines (anti blocking coatings, scratch resistant coatings on plastic parts). ii) Lubricant – free bearings.
2. **Electronic industry:** Data memory (MRAM,GMR-HD), Displays(OLED,FED), Laser diodes, Glass fibres
3. **Automotive industry:** Light weight construction, Painting (fillers, base coat, clear coat), Sensors, Coating for wind screen and car bodies.
4. **Construction:** Construction materials, Thermal insulation, Flame retardants.
5. **Chemical industry:** Fillers for painting systems, Coating systems based on nano Composites. Impregnation of papers, Magnetic Fluids.
6. **Medicine:** Drug delivery systems, Agents in cancer therapy, Anti microbial agents And coatings, Medical rapid tests Active agents.
7. **Energy:** Fuel cells, Solar cells, batteries, Capacitors.
8. **Cosmetics:** Sun protection, Skin creams, Tooth paste, Lipsticks.

Short Questions

1. What are nano materials?
2. Write about origin of nanotechnology.
3. Why do nanomaterials exhibit different properties?
4. Explain about surface area to volume ratio.
5. What is meant by quantum confinement?
6. Discuss about sol-gel technique.
7. Give the applications of nanotechnology.
8. Write the advantages of TEM.
9. What are the applications of XRD?

Long Questions

1. Explain the fabrication of nanoparticles by CVD method.
2. Explain the fabrication of nanoparticles by Sol-Gel.
3. Explain the fabrication of nanoparticles by PVD method.
4. Discuss about TEM technique to characterize nanoparticles.
5. Explain how SEM can be used to characterize nanoparticles.
6. Discuss about XRD technique to characterize nanoparticles